



Electrical and Computer Engineering
University of Thessaly

ECE214 - Introduction to Electronics

Analysis of nMOS and pMOS Transistor Characteristics

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Final Project Report
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1 Introduction

The objective of this laboratory exercise is the detailed characterization of Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs). Specifically, we analyze the I_{DS} vs. V_{DS} characteristics for both nMOS and pMOS devices across three manufacturing technologies: $0.35\ \mu\text{m}$, $0.25\ \mu\text{m}$, and $180\ \text{nm}$. The study evaluates the impact of technology scaling and temperature variations (27°C and 70°C) on device performance.

2 Theoretical Background

The MOSFET operates in three distinct regions determined by the gate-source (V_{GS}) and drain-source (V_{DS}) voltages:

- **Cut-off Region** ($V_{GS} < V_{th}$): The transistor is OFF, and $I_{DS} \approx 0$.
- **Linear (Triode) Region** ($V_{GS} > V_{th}$ and $V_{DS} < V_{GS} - V_{th}$): The channel acts as a voltage-controlled resistor.
- **Saturation Region** ($V_{GS} > V_{th}$ and $V_{DS} \geq V_{GS} - V_{th}$): Ideally, I_{DS} becomes independent of V_{DS} .

Second-order effects analyzed in this report include:

1. **Channel Length Modulation:** The decrease in effective channel length as V_{DS} increases, resulting in a positive slope in the saturation region.
2. **Velocity Saturation:** A phenomenon prevalent in short-channel devices where carrier velocity reaches a limit, leading to a linear rather than quadratic relationship between I_{DS} and V_{GS} .

3 Experimental Results and Analysis

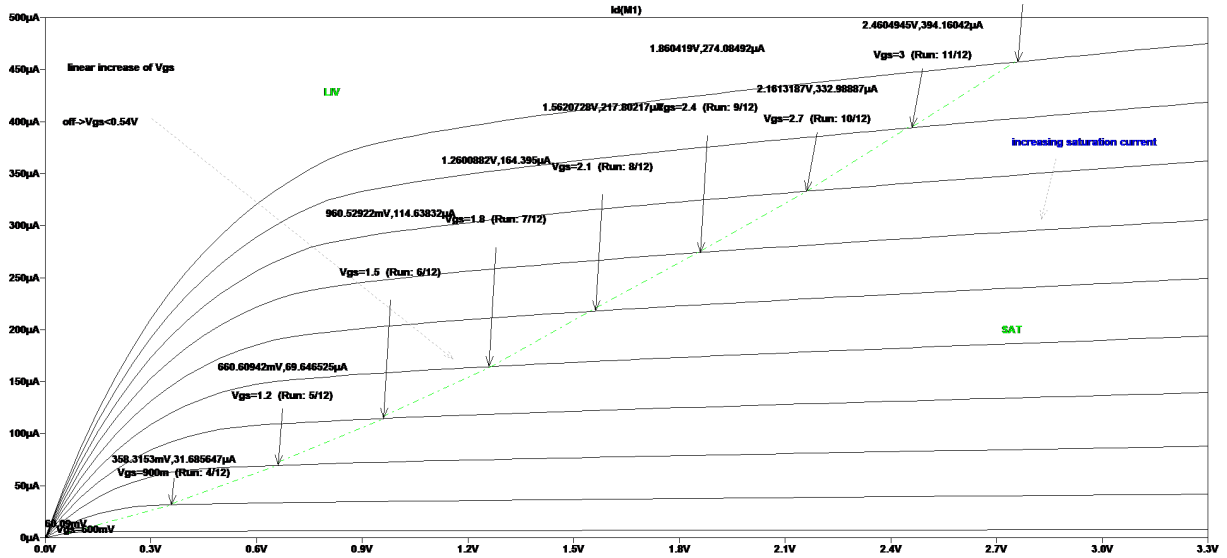
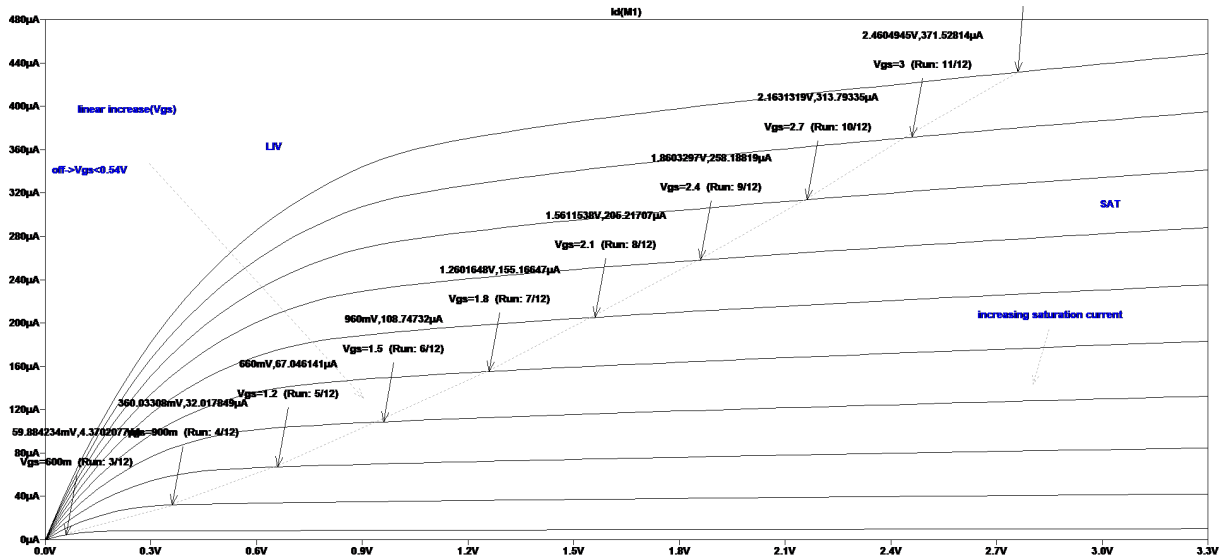
In this section, we present the simulation results for the I-V characteristics of all three technology nodes ($0.35\ \mu\text{m}$, $0.25\ \mu\text{m}$, and $180\ \text{nm}$). The analysis focuses on characterizing the performance of nMOS and pMOS transistors, evaluating their behavior under different thermal conditions at 27°C and 70°C . Our study aims to identify key operational parameters and observe the transition from long-channel to short-channel phenomena.

3.1 Analysis of nMOS Technologies

The starting point of this project is the examination of the I-V characteristics for the long-channel nodes under both available temperatures in order to evaluate mobility degradation and channel length modulation.

3.1.1 TSMC $0.35\ \mu\text{m}$ Characteristics

Examining the I-V characteristic plots (Figures 1 and 2) obtained from the LTspice simulations, we first identify the saturation points for each curve. These points are determined by the condition $V_{DS,sat} = V_{GS} - V_{th}$. The threshold voltage (V_{th}) is defined according to the specific

Figure 1: nMOS 0.35 μm I-V Characteristics at 27°C.Figure 2: nMOS 0.35 μm I-V Characteristics at 70°C.

technology node and transistor type (nMOS/pMOS), as specified by the VTO parameter in the embedded SPICE model files.

By connecting these saturation points, we define the boundary (represented by the dashed line in the plots) that separates the operational plane into two distinct regions. Any point on the I-V characteristics located to the left of this boundary belongs to the Linear (Triode) region, while points located to the right reside in the Saturation region.

At these nodes, the primary non-ideal effect observed is Channel Length Modulation, which creates a non-zero slope in the saturation region. This indicates that the drain current I_D is not entirely independent of the voltage V_{DS} ; on the contrary, it appears to increase slightly as V_{DS} values rise. Furthermore, it can be observed that the relationship between V_{GS} and I_D is not quadratic but linear. Since every constant increase in the V_{GS} value results in a proportional increase in the current I_{DS} , the gaps between consecutive I-V curves remain equal. This is a clear indication of the Velocity Saturation phenomenon.

The transition to 70°C does not show any significant difference from the previous observa-

tions under 27°C so no further analysis is required.

3.1.2 TSMC 0.25 μm Characteristics

Similarly to the previous simulations, but using fewer V_{GS} steps due to the reduced supply voltage ($V_{DD} = 2.5V$), we observe the same general behavior in the I-V characteristics for the 0.25 μm technology across both temperatures. The corresponding simulation plots for this particular technology can be observed below in Figures 3 and 4.

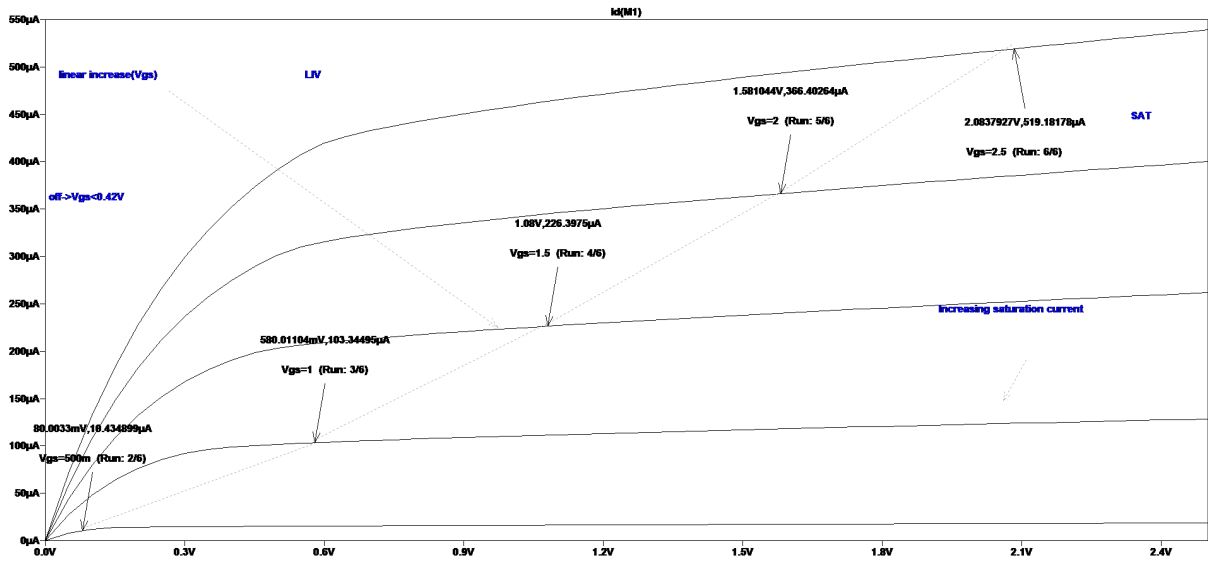


Figure 3: nMOS 0.25 μm I-V Characteristics at 27°C.

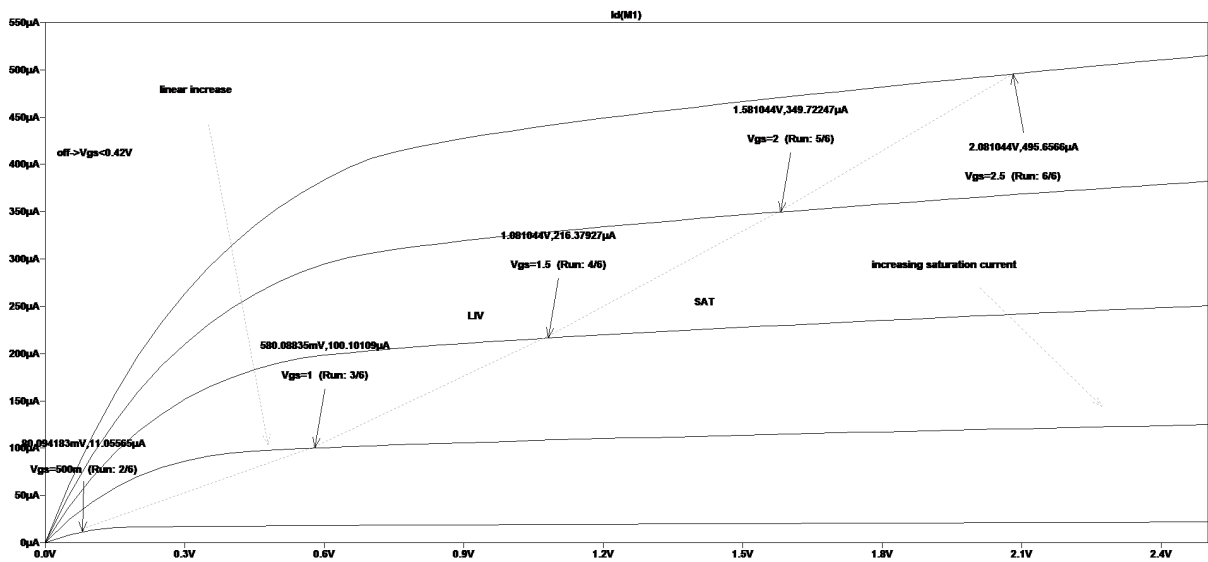


Figure 4: nMOS 0.25 μm I-V Characteristics at 70°C.

3.1.3 TSMC 180 nm Characteristics

We notice a similar pattern to the previous technologies, albeit with some significant differences. Firstly, the dependence of I_D on V_{DS} is less pronounced than in the previous paradigms,

with the slope in the saturation region being almost zero. This implies that the Channel Length Modulation phenomenon is less intense in this specific technology node. Another point that merits consideration is the significant increase in the current values (I_D) for the same V_{GS} and V_{DS} levels compared to the larger technology nodes. Lastly, the number of I-V characteristic curves is reduced due to the larger step size implemented to match the lower V_{DD} of the 180 nm process. Below we can observe the plots obtained from the spice simulations (5 & 6)

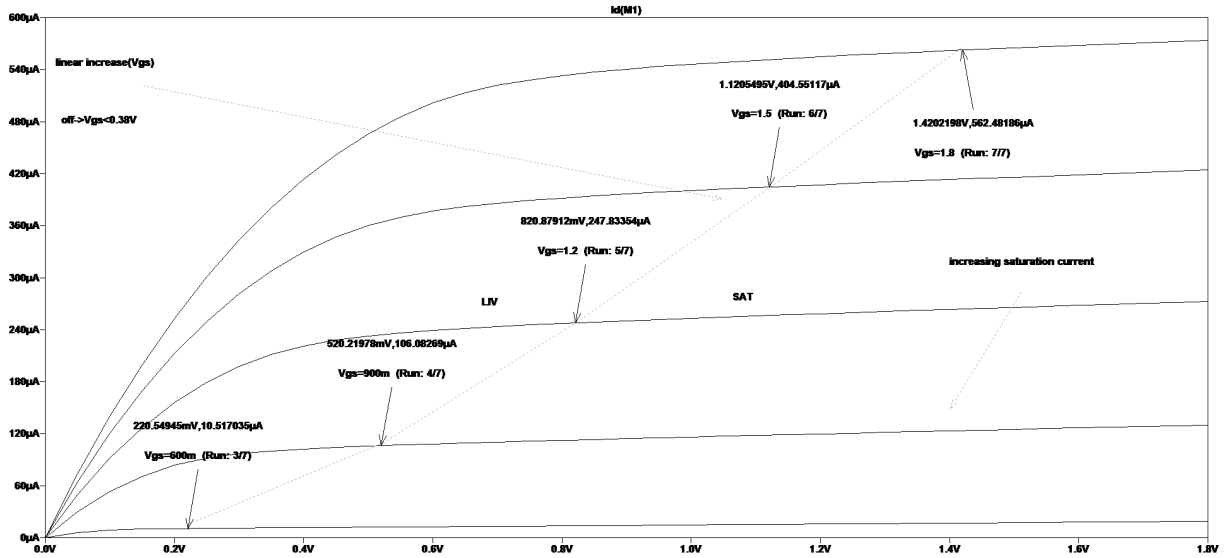


Figure 5: nMOS 0.25 μm I-V Characteristics at 27°C.

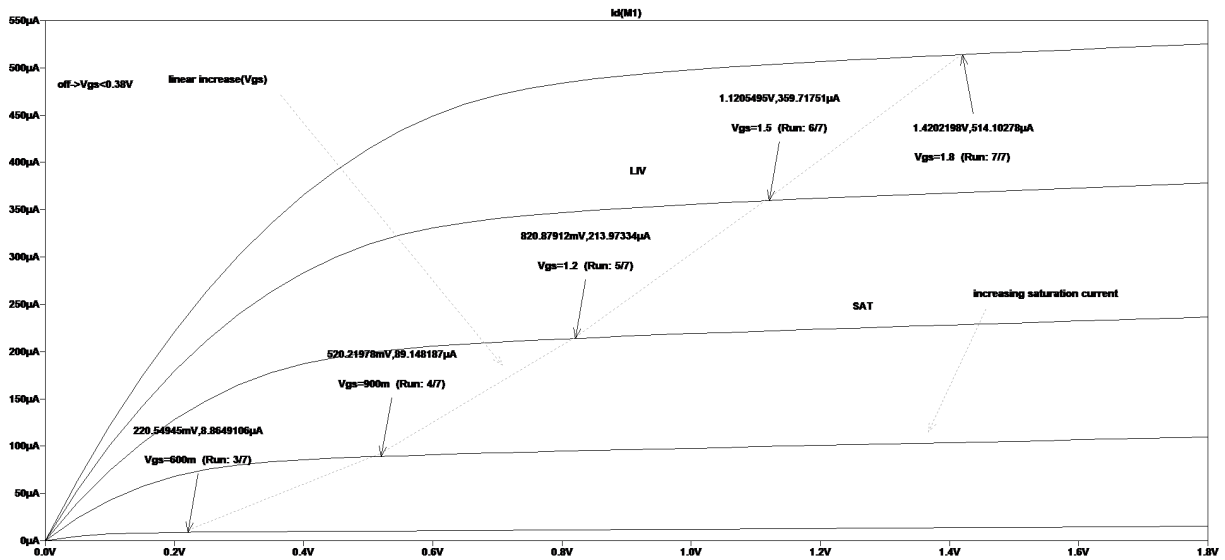


Figure 6: nMOS 0.25 μm I-V Characteristics at 70°C.

As noted in the simulation, the I_{DS} curves for equal increments of V_{GS} are nearly equally spaced. This linear dependence of I_{DS} on V_{GS} (instead of quadratic) is the definitive signature of Velocity Saturation.

3.2 Analysis of pMOS Technologies

Following the same methodology for pMOS transistors, we observe similar physical phenomena. The values of V_{GS} and V_{DS} are negative and the current flows in the opposite direction.

We similarly calculate the saturation points for every V_{GS} curve for this particular technology and construct the boundary line that divides the operating plane into two regions. In this case, however, due to the negative voltage polarities of the pMOS transistor, the saturation region is located on the left semi-plane while the linear region is on the right.

3.2.1 TSMC 0.35 μm Characteristics

According to the simulation results (Figures 7 and 8), the Channel Length Modulation phenomenon is more prominent in pMOS compared to nMOS transistors, as the magnitude of the drain current continues to increase noticeably even after crossing the saturation point. Furthermore, the vertical gaps between the characteristic curves for different values of V_{GS} differ slightly from one to another. This suggests that while the Velocity Saturation phenomenon is still present, it is not as intense in pMOS devices as it is in their nMOS counterparts, primarily due to the lower mobility of holes.

Furthermore, a consistent observation across all pMOS characteristics, regardless of the channel length, is that the drain current magnitude increases as the operating temperature decreases. This phenomenon is attributed to the enhancement of carrier mobility (μ) at lower temperatures, which allows for higher current conduction for the same bias conditions.

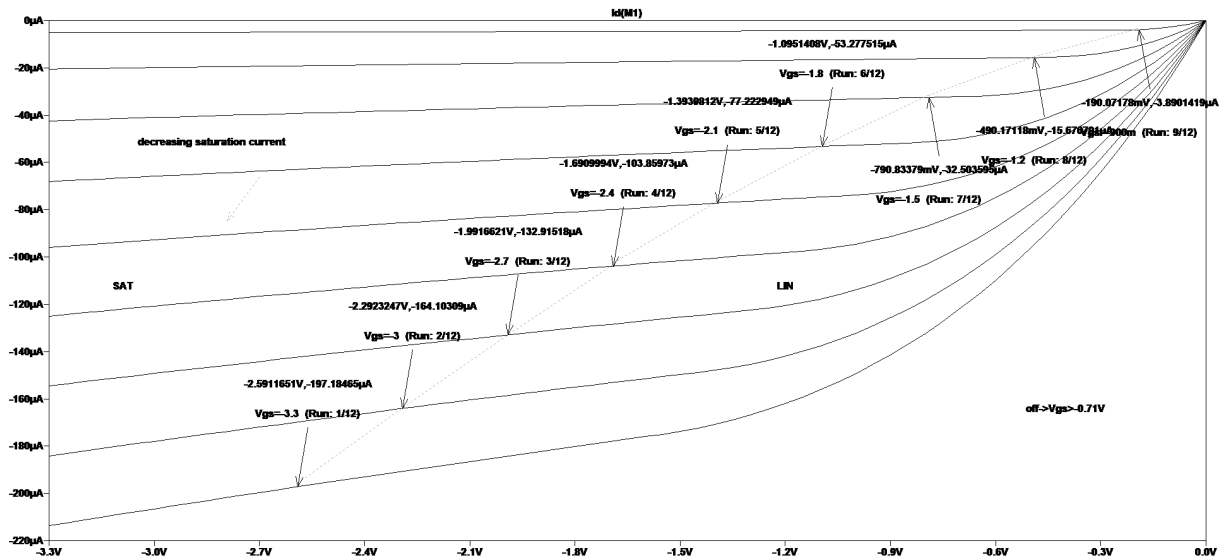
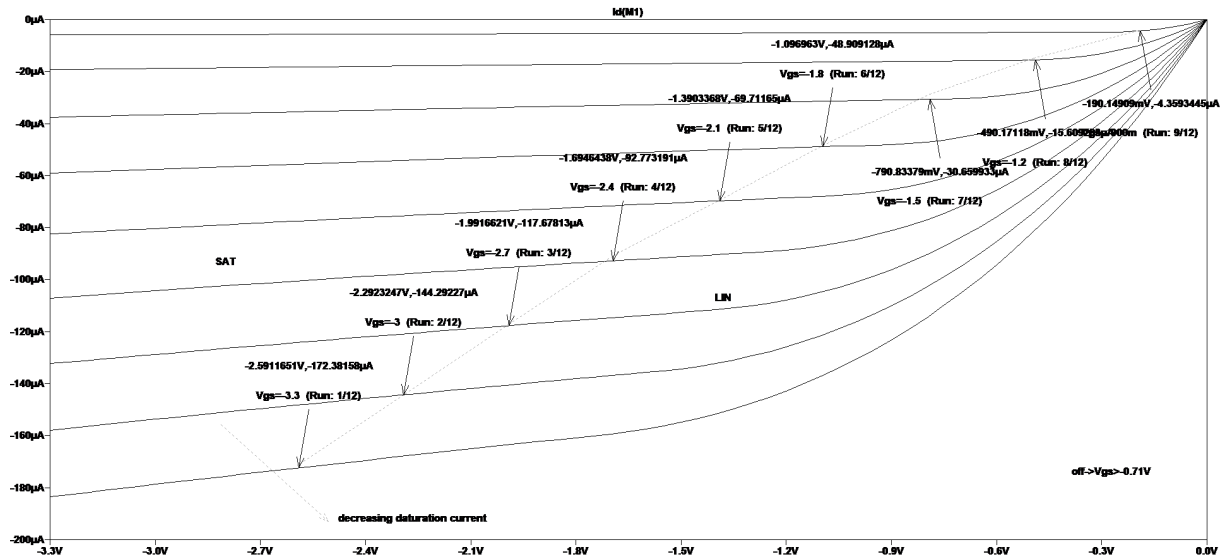
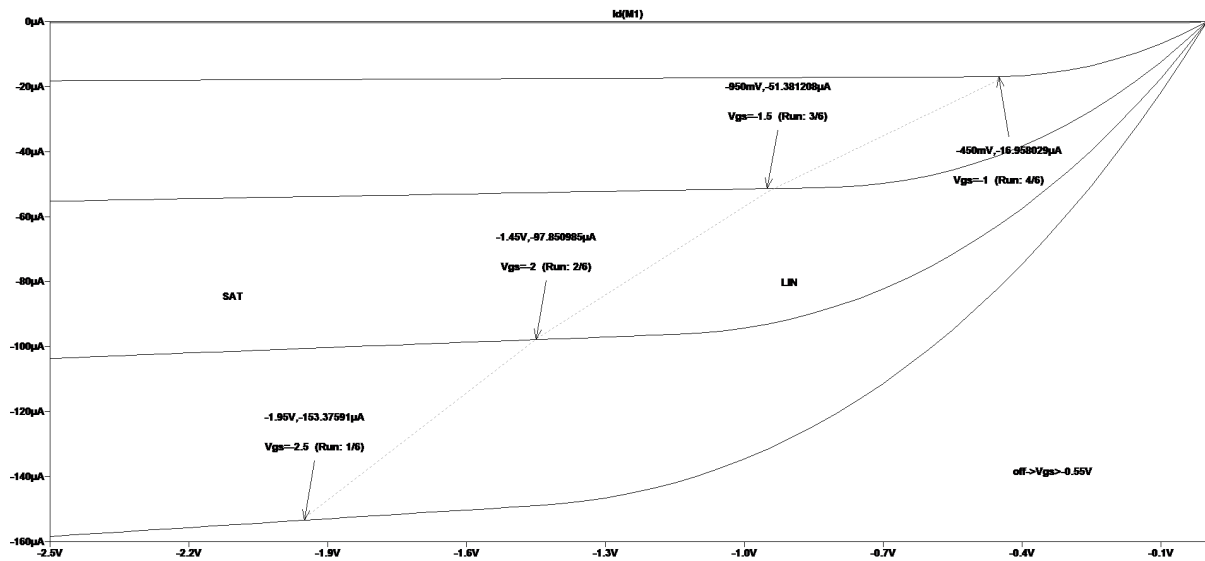


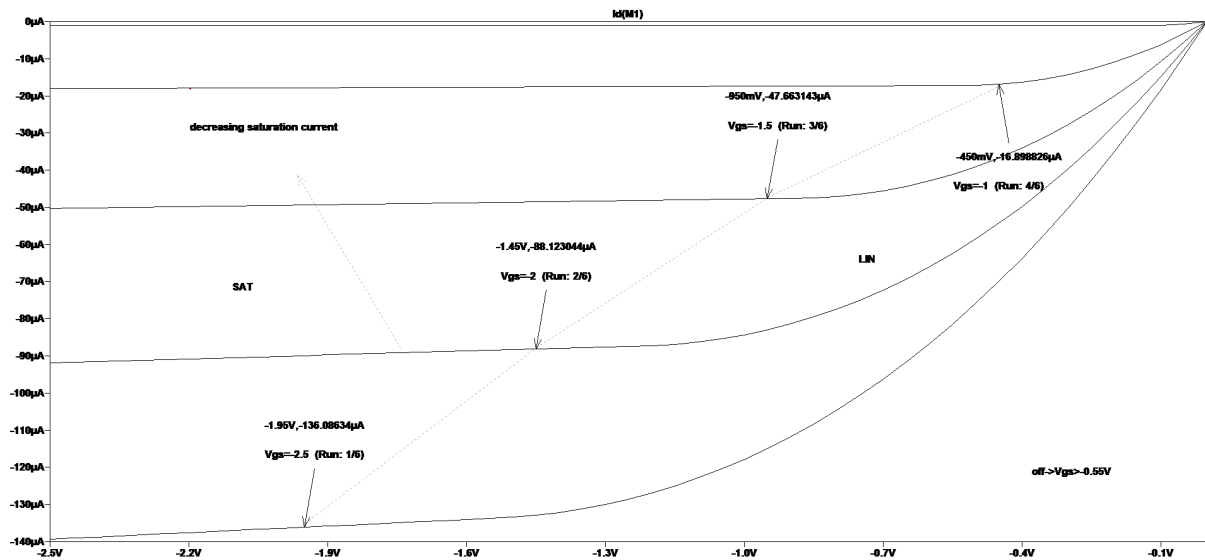
Figure 7: pMOS 0.35 μm I-V Characteristics at 27°C.

Figure 8: pMOS 0.35 μm I-V Characteristics at 70°C.

3.2.2 TSMC 0.25 μm Characteristics

The exact same observations can be made regarding the pMOS transistors with a channel length of 0.25 μm under both 27°C and 70°C thermal conditions, as illustrated in Figures 9 and 10.

Figure 9: pMOS 0.25 μm I-V Characteristics at 27°C.

Figure 10: pMOS 0.25 μm I-V Characteristics at 70°C.

3.2.3 TSMC 180 nm Characteristics

The exact same phenomena analysed in the previous technology nodes are valid for this one as well, requiring no further analysis. The consistency of these results across the 0.35 μm , 0.25 μm , and 180 nm nodes reinforces the observed trends regarding temperature effects and short-channel behaviour. The spice simulation results for this technology are illustrated below (11 & 12)

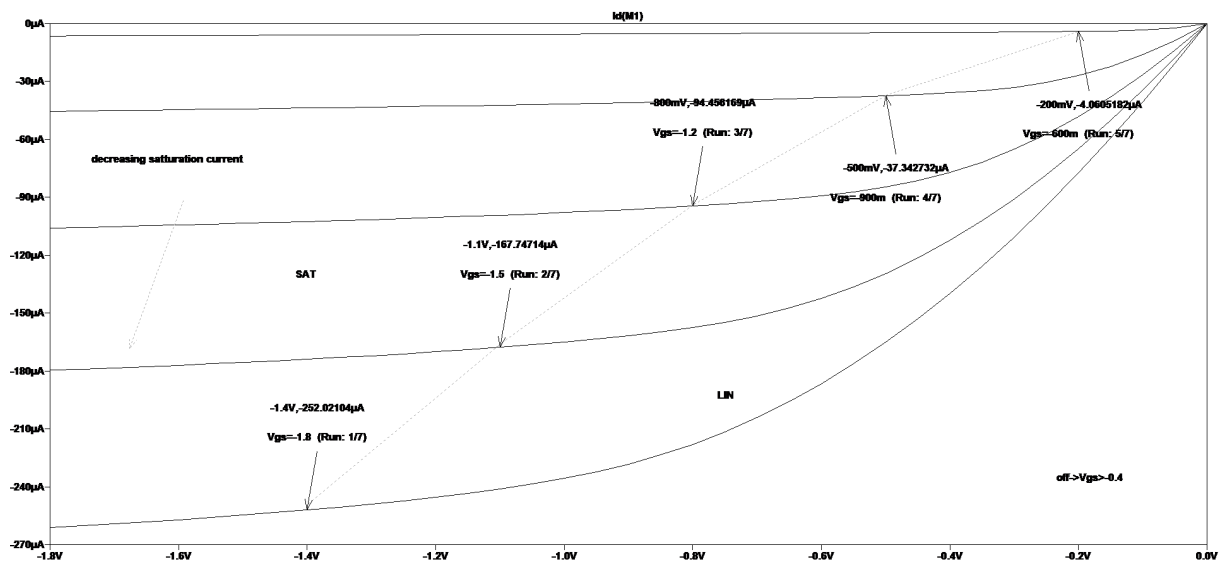


Figure 11: pMOS 180 nm I-V Characteristics at 27°C.

4 Conclusion

In this laboratory project, we conducted a detailed characterization of nMOS and pMOS transistors across three distinct technology nodes: 0.35 μm , 0.25 μm , and 180 nm. By analyzing the

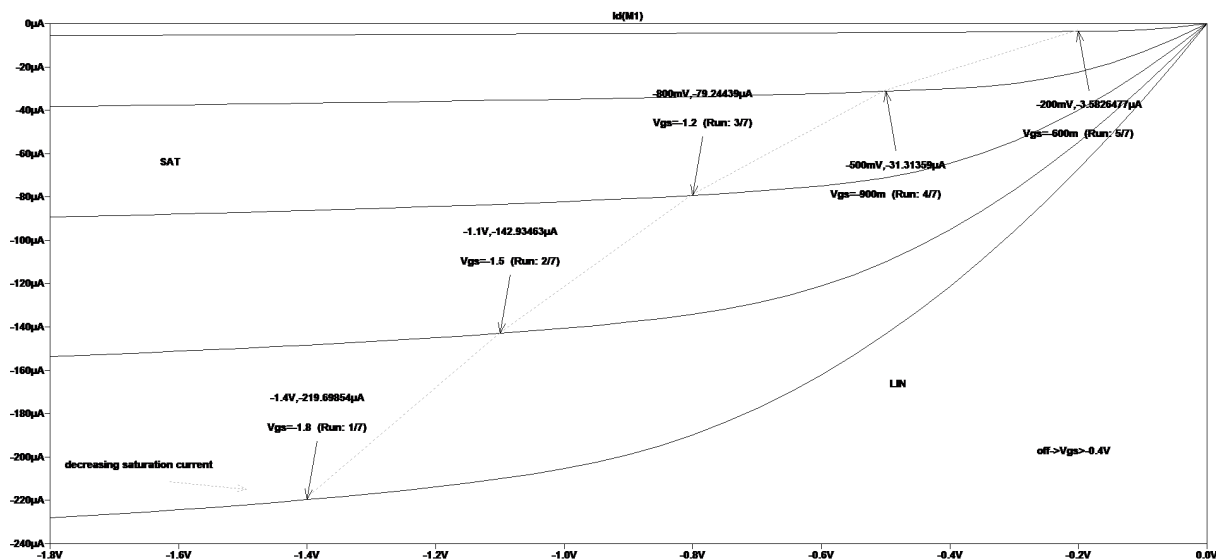


Figure 12: pMOS 180 nm I-V Characteristics at 70°C.

I-V characteristics under different thermal conditions (27°C and 70°C), several key conclusions were drawn:

- **Thermal Effects:** Across all technologies and transistor types, an increase in operating temperature resulted in a significant reduction of the drain current (I_D). This confirms that carrier mobility (μ) degradation is the dominant factor at higher temperatures.
- **Channel Length Modulation (CLM):** We observed that as the technology scales, CLM remains a persistent non-ideal effect. However, its impact was found to be more prominent in pMOS transistors compared to their nMOS counterparts, evidenced by the more significant slope in the saturation region.
- **Velocity Saturation:** The linear spacing between I-V curves for constant V_{GS} increments provided definitive evidence of velocity saturation, an effect that was particularly intense in nMOS devices.
- **Technology Scaling:** The simulations successfully demonstrated the trade-offs of scaling, such as the reduction of supply voltage (V_{DD}) and the resulting changes in current magnitude and transistor efficiency.

In summary, the project provided a practical validation of semiconductor theory, highlighting how physical dimensions and environmental factors like temperature fundamentally alter the performance of modern integrated circuits.